Apex Global Bank

Quantitative Research & Treasury

Bank Balance Sheet Optimization

Theory, Mathematics, Regulatory Framework,
Technology, and Revenue Generation

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This document is proprietary to Apex Global Bank. It describes the quantitative framework, regulatory constraints, computational infrastructure, and revenue-generation mechanisms underlying the bank's balance sheet optimization program. Sections 3–8 contain material non-public information. Distribution restricted to authorized personnel.

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1 Introduction and Strategic Context

1.1 What Is Balance Sheet Optimization?

A bank's balance sheet is fundamentally a portfolio of financial claims. On the asset side: loans, securities, derivatives, and cash. On the liability side: deposits, wholesale funding, issued debt, and equity. *Balance sheet optimization* is the discipline of constructing and managing this portfolio to maximize risk-adjusted return subject to regulatory constraints, liquidity requirements, and the bank's own risk appetite.

Unlike trading book optimization, which operates on a horizon of hours to days, balance sheet optimization operates on horizons of months to years. It is a strategic planning function that touches every product line: commercial lending, retail deposits, fixed-income securities portfolios, derivatives books, repo/securities finance, and off-balance-sheet vehicles.

At the highest level, the problem is:

$$\max_{\mathbf{a},\mathbf{l}} \quad \text{RORWA}(\mathbf{a},\mathbf{l}) \quad \text{subject to} \quad \mathbf{g}(\mathbf{a},\mathbf{l}) \leq \mathbf{0} \tag{1}$$

where $\mathbf{a}$ is the asset allocation vector, $\mathbf{l}$ is the liability structure vector, RORWA is the return on risk-weighted assets, and $\mathbf{g}$ encodes the regulatory and risk constraints (capital ratios, liquidity ratios, concentration limits, stress test floors).

In practice this is far more complex than a single optimization: the constraints are multi-dimensional, some are non-linear, many are path-dependent over time, and the objective itself is a function of models that embed their own uncertainty.

1.2 Why It Matters: Revenue at Stake

At a JPMorgan-scale institution ($3.9 trillion balance sheet), a 5 basis point improvement in return on assets (ROA) is worth approximately $2 billion per year. The levers available through balance sheet optimization include:

- **Capital efficiency**: replacing high-RWA assets with equivalent-yield lower-RWA assets releases CET1 that can be redeployed or returned to shareholders via buy-backs.
- **Funding cost reduction**: optimizing the liability mix (longer-term funding when the curve is inverted; shorter-term when normally sloped) reduces the weighted average cost of funds.
- **NII maximization**: positioning the balance sheet's asset-liability duration gap to benefit from expected rate moves.
- **Regulatory arbitrage** (within the rules): choosing product structures that achieve the same economic exposure with lower capital consumption (e.g., guarantees vs. direct loans, cleared vs. bilateral derivatives).
- **Liquidity buffer efficiency**: minimizing the drag from mandatory HQLA holdings while maintaining LCR/NSFR compliance.

1.3 The Regulatory Stack

Balance sheet optimization operates inside a dense regulatory framework. The binding constraints are:

Table 1: Key Regulatory Constraints on the Bank Balance Sheet

Regulation	Metric	Minimum	Apex Current
Basel III / 12 CFR 3	CET1 ratio	4.5% (+ CCB 2.5%)	13.0%
Basel III / 12 CFR 3	Tier 1 ratio	6.0% (+ CCB)	14.5%
Basel III / 12 CFR 3	Total Capital ratio	8.0% (+ CCB)	16.2%
Basel III / 12 CFR 3	Leverage ratio	3.0% (SLR)	7.8%
Basel III / 12 CFR 249	LCR	100%	118%
Basel III / 12 CFR 249	NSFR	100%	112%
Dodd-Frank / 12 CFR 252	DFAST CET1 floor (adverse)	4.5% (regulatory)	8.7% trough
Basel IV / FRTB	FRTB-SA floor	Binding from 2025	—
Concentration rules	Single-name limit	25% of CET1	Monitored

Each of these constraints reduces the feasible set and potentially creates binding limits that shape the optimal balance sheet structure.

2 The Optimization Problem: Mathematical Formulation

2.1 State Variables and Decision Variables

Let $t = 0, 1, \ldots, T$ index quarterly periods. The state of the balance sheet at time t is described by:

- $A_k(t)$: balance in asset class k (loans, AFS securities, HTM securities, derivatives, cash, other), for $k = 1, \ldots, K_A$
- $L_j(t)$: balance in liability class j (demand deposits, time deposits, FHLB advances, senior unsecured debt, repo, equity), for $j = 1, \ldots, K_L$
- $r_k^A(t)$: yield on asset class k at time t
- $r_j^L(t)$: cost of liability class j at time t
- $\rho_k(t)$: risk weight for asset class k (SA or IMA)
- q_t: macroeconomic state vector (GDP growth, unemployment, rates)

The decision variables are the *incremental* changes:

$$\Delta A_k(t), \quad \Delta L_j(t), \quad \forall\, k, j, t \tag{2}$$

subject to the balance sheet identity:

$$\sum_k A_k(t) = \sum_j L_j(t), \qquad \forall\, t \tag{3}$$

2.2 Objective Function

The primary objective is to maximize the net present value of risk-adjusted returns over the planning horizon $[0, T]$:

$$\max_{\{\Delta A_k(t),\, \Delta L_j(t)\}} \sum_{t=0}^{T} \delta^t \left[\mathrm{NII}(t) + \mathrm{Fee}(t) - \mathrm{Credit_Loss}(t) - \mathrm{OpRisk}(t) - \mathrm{FTP_Cost}(t)\right] \quad (4)$$

where $\delta = 1/(1+h)$ is the discount factor (hurdle rate h is the bank's cost of equity, typically CAPM-derived: $h = r_f + \beta(r_m - r_f)$).

Equivalently, using the RORWA metric preferred by bank management:

$$\mathrm{RORWA}(t) = \frac{\mathrm{Net_Income_after_tax}(t)}{\mathrm{RWA}(t)} \quad (5)$$

2.3 Net Interest Income

$$\mathrm{NII}(t) = \sum_k r_k^A(t) \cdot A_k(t) - \sum_j r_j^L(t) \cdot L_j(t) \quad (6)$$

The net interest margin (NIM) is:

$$\mathrm{NIM}(t) = \frac{\mathrm{NII}(t)}{\mathrm{Average_Earning_Assets}(t)} \quad (7)$$

2.4 Capital Constraints

The CET1 constraint:

$$\frac{\mathrm{CET1}(t)}{\mathrm{RWA}(t)} \geq \kappa_{\mathrm{CET1}}, \qquad \forall\, t \quad (8)$$

where $\kappa_{\mathrm{CET1}} = 7\%$ (4.5% minimum + 2.5% CCB) in normal conditions and $\kappa_{\mathrm{CET1}}^{\mathrm{stress}} = 6.5\%$ under severely adverse DFAST.

Risk-weighted assets:

$$\mathrm{RWA}(t) = \sum_k \rho_k(t) \cdot A_k(t) \quad (9)$$

The leverage ratio constraint:

$$\frac{\mathrm{Tier1}(t)}{\mathrm{Total_Leverage_Exposure}(t)} \geq \kappa_{\mathrm{SLR}} \quad (10)$$

where Total Leverage Exposure = total assets + off-balance-sheet items + derivative add-ons. The SLR (Supplementary Leverage Ratio) is often the binding constraint for large banks with low-RWA assets (Treasuries, Agency MBS).

2.5 Liquidity Constraints

LCR (Liquidity Coverage Ratio):

$$\mathrm{LCR}(t) = \frac{\mathrm{HQLA}(t)}{\mathrm{Net_Cash_Outflows}_{30d}(t)} \geq 1.0 \tag{11}$$

where HQLA (High-Quality Liquid Assets) = Level 1 assets (cash, central bank reserves, Treasuries) at 100% + Level 2A assets (Agency debt, covered bonds) at 85% + Level 2B assets (non-financial corporate bonds, equities) at 50%.

Net cash outflows are computed from stressed runoff rates applied to each liability category:

$$\mathrm{Net_Cash_Outflows}_{30d}(t) = \sum_j \alpha_j \cdot L_j(t) - \mathrm{Cash_Inflows}(t) \tag{12}$$

where α_j is the 30-day stressed runoff rate for liability class j (e.g., $\alpha_{\text{retail deposits}} = 3\%$, $\alpha_{\text{wholesale unsecured}} = 25\%$, $\alpha_{\text{secured funding}} = 0\%$ to 25% depending on collateral quality).

NSFR (Net Stable Funding Ratio):

$$\mathrm{NSFR}(t) = \frac{\mathrm{ASF}(t)}{\mathrm{RSF}(t)} \geq 1.0 \tag{13}$$

Available Stable Funding (ASF):

$$\mathrm{ASF}(t) = \sum_j \beta_j^{\mathrm{ASF}} \cdot L_j(t) \tag{14}$$

Required Stable Funding (RSF):

$$\mathrm{RSF}(t) = \sum_k \beta_k^{\mathrm{RSF}} \cdot A_k(t) \tag{15}$$

where β^{ASF} and β^{RSF} are Basel-prescribed factors. High-RSF assets (e.g., loans ¿ 1 year, RSF = 65–85%) paired with low-ASF short-term funding creates NSFR pressure.

2.6 DFAST Stress Constraint

The stress test constraint ensures the CET1 ratio survives the 9-quarter projection even under the severely adverse scenario:

$$\min_{t \in [1,9]} \mathrm{CET1_ratio}^{\mathrm{sev_adv}}(t) \geq \kappa_{\mathrm{floor}} \tag{16}$$

where $\kappa_{\mathrm{floor}} = 6.5\%$ (internal risk appetite) or 4.5% (regulatory minimum).

This constraint is forward-looking and non-linear: it couples the current balance sheet composition to the modelled stressed loss path over 9 quarters.

2.7 Concentration Constraints

Single-name concentration:

$$A_{\text{counterparty-}i}(t) \leq 0.25 \cdot \text{CET1}(t), \qquad \forall\, i \tag{17}$$

Sector concentration:

$$\sum_{i \in \text{sector-}s} A_i(t) \leq 0.25 \cdot \text{Total_Assets}(t), \qquad \forall\, s \tag{18}$$

2.8 The Full Constrained Problem

Combining constraints (3)–(17):

$$\begin{aligned}
\max_{\Delta A, \Delta L} \quad & \sum_{t=0}^{T} \delta^t \left[\text{NII}(t) - \text{CreditLoss}(t) - \text{OpRisk}(t)\right] \\
\text{s.t.} \quad & (3) : \text{Balance sheet identity} \\
& (8) : \text{CET1 ratio} \geq 7\% \\
& (10) : \text{SLR} \geq 3\% \\
& (11) : \text{LCR} \geq 100\% \\
& (13) : \text{NSFR} \geq 100\% \\
& (16) : \text{DFAST CET1 trough} \geq 6.5\% \\
& (17) : \text{Concentration limits} \\
& A_k(t) \geq 0, \quad L_j(t) \geq 0 \quad \forall\, k, j, t
\end{aligned} \tag{19}$$

This is a multi-period nonlinear program (NLP) with integer components (lending decisions), stochastic elements (credit losses depend on macro realization), and regulatory feedback (RWA changes with asset mix). In practice it is decomposed into subproblems: capital optimization, liquidity optimization, NII/duration management, and pricing.

3 Capital Optimization

3.1 RWA Minimization and the Capital Efficiency Frontier

The *capital efficiency frontier* maps every unit of expected return against the required capital. The Sharpe-style metric is:

$$\text{RAROC}(k) = \frac{r_k^A - r_k^{\text{FTP}} - \text{EL}_k}{\text{EC}_k} \tag{20}$$

where:

- r_k^A: gross yield on asset class k
- r_k^{FTP}: matched-maturity FTP charge (Section 6)
- $\text{EL}_k = \text{PD}_k \cdot \text{LGD}_k$: expected loss rate
- EC_k: economic capital allocation per unit (see below)

A business line is value-creating if $\text{RAROC}(k) > h$ (hurdle rate). The optimization reallocates the balance sheet toward high-RAROC activities.

3.2 Economic Capital Allocation

Regulatory capital (RWA-based) and economic capital (model-based) can diverge. Economic capital is typically computed using a portfolio credit VaR at a 99.9% confidence level (consistent with BBB-rated survival probability):

$$\mathrm{EC}_{\mathrm{portfolio}} = \mathrm{VaR}_{99.9\%}(\mathrm{CreditLoss}_{1yr}) - \mathrm{EL} \tag{21}$$

Per-obligor capital is allocated using the marginal contribution:

$$\mathrm{EC}_i = \frac{\partial \mathrm{EC}_{\mathrm{portfolio}}}{\partial w_i} = \mathrm{Corr}(L_i, L_{\mathrm{portfolio}}) \cdot \sigma_{L_i} \cdot z_{99.9\%} \tag{22}$$

where L_i is the loss random variable for obligor i. This "marginal EC" framework means that adding a low-correlation obligor consumes less capital than an equally risky but highly correlated one.

3.3 SA vs. IMA Capital Arbitrage

Under Basel III, banks with IMA approval compute market risk capital using internal VaR models (lower capital for diversified portfolios). Under FRTB, all banks report SA capital as a floor.

The capital ratio under SA vs. IMA:

$$\mathrm{RWA}^{\mathrm{SA}}(k) = \rho_k^{\mathrm{SA}} \cdot \mathrm{EAD}_k \tag{23}$$

$$\mathrm{RWA}^{\mathrm{IMA}}(k) = 12.5 \cdot \left[\max\left(\mathrm{VaR}_t, \bar{k} \cdot \overline{\mathrm{VaR}}_{60d}\right) + \max\left(\mathrm{SVaR}_t, \bar{k} \cdot \overline{\mathrm{SVaR}}_{60d}\right)\right] \tag{24}$$

For a well-diversified trading book, $\mathrm{RWA}^{\mathrm{IMA}} < \mathrm{RWA}^{\mathrm{SA}}$ by 30–60%, representing a significant capital saving that justifies the cost of IMA model development and validation.

3.4 Capital Buffer Tiering

Optimal capital management requires maintaining a buffer above the regulatory minimum. The buffer serves multiple purposes:

$$\mathrm{CET1}_{\mathrm{target}} = \kappa_{\min} + B_{\mathrm{CCB}} + B_{\mathrm{DFAST}} + B_{\mathrm{management}} \tag{25}$$

- $\kappa_{\min} = 4.5\%$: Basel III minimum
- $B_{\mathrm{CCB}} = 2.5\%$: Capital Conservation Buffer
- B_{DFAST}: buffer between trough (severely adverse) and minimum; at Apex this is $\approx 2.2\%$ (trough 6.7%, minimum 4.5%)
- $B_{\mathrm{management}} = 1.0$–$1.5\%$: management buffer for strategic flexibility, acquisitions, unexpected losses

A bank holding excess capital beyond these buffers faces pressure from shareholders to deploy it efficiently via buybacks, dividends, or accretive acquisitions.

4 Liquidity Optimization

4.1 The Liquidity Drag and HQLA Portfolio Management

LCR and NSFR compliance requires the bank to hold a portfolio of High-Quality Liquid Assets (HQLA). HQLA earns below-market yields (e.g., Treasuries at 4.5% vs. corporate loans at 6.5%+), creating a structural "liquidity drag" on NIM:

$$\text{Liquidity_Drag} = (\bar{r}^A - r^{\text{HQLA}}) \cdot \frac{\text{HQLA}}{A} \tag{26}$$

At Apex with HQLA of $45B and a balance sheet of $285B (16% HQLA ratio):

$$\text{Liquidity_Drag} \approx (6.2\% - 4.5\%) \times 16\% = 27\,\text{bps NIM} \tag{27}$$

HQLA portfolio optimization seeks to maximize yield within the HQLA composition constraints:

$$\max_{\text{HQLA mix}} \sum_{i \in \text{HQLA}} w_i \cdot r_i \quad \text{s.t.} \quad \begin{cases} \text{Level 1} \geq 60\% \text{ of HQLA} \\ \text{Level 2A} \leq 40\% \text{ of HQLA} \\ \text{Level 2B} \leq 15\% \text{ of HQLA} \\ \text{Duration} \leq D_{\max} \\ \text{LCR} \geq 110\% \text{ (internal buffer)} \end{cases} \tag{28}$$

Within Level 1, the bank can substitute between cash (0% yield), central bank reserves (Fed Funds: 5.25%), and Treasury bills/bonds (4.5%–4.7%). The optimization shifts toward Treasuries and away from excess cash.

4.2 Intraday Liquidity Management

Intraday liquidity management is distinct from LCR/NSFR. The bank must fund settlement obligations throughout the day via Fedwire and CHIPS:

$$\text{Intraday_Peak}(t) = \max_{\tau \in [0, T_{\text{day}}]} [\text{Outflows}(\tau) - \text{Inflows}(\tau) - \text{Reserve}(\tau)]^+ \tag{29}$$

Minimizing the peak intraday liquidity requirement (and thus the cost of intraday credit from the Fed) is a treasury optimization problem. Tools include:

- Prefunding (posting collateral to Fed discount window)
- Payment sequencing (delaying non-urgent payments to afternoon)
- Correspondent banking netting

4.3 Funding Mix Optimization

The liability-side optimization seeks the lowest-cost, most stable funding mix consistent with NSFR compliance:

$$\min_{\Delta L_j} \sum_j r_j^L \cdot L_j \quad \text{s.t.} \quad \text{NSFR} \geq 100\%, \quad \text{LCR} \geq 100\% \tag{30}$$

The trade-off: demand deposits are cheap (near-zero rate) but have high NSFR ASF factors only if stable (50% core vs. 0% non-stable). Long-term unsecured debt has a high ASF (100%) but is expensive (e.g., OIS + 80–120bps for senior unsecured).

Definition 1 (Optimal Funding Tenor)**.** *Given a flat (or inverted) yield curve, the optimal funding maturity τ^* for a T-year asset satisfies:*

$$\tau^* = \arg\min_{\tau} \left[r^L(\tau) + \mathit{Rollover_Risk}(\tau, T) \right] \tag{31}$$

where $\mathit{Rollover_Risk}(\tau, T) = \sigma_r^2 \cdot (T - \tau)$ is the expected cost of refunding if short-term rates rise.

In a normal upward-sloping curve, short-term funding is cheaper but rollover risk is higher. In an inverted curve (2022–2024), long-term funding may be cheaper even without considering rollover risk.

5 Asset-Liability Management (ALM)

5.1 NII Sensitivity and the Repricing Gap

The repricing gap model described in Section 2 provides a first-order linear approximation of NII sensitivity:

$$\Delta\text{NII} \approx \text{RSA}^{12m} \cdot \Delta r - \text{RSL}^{12m} \cdot \Delta r = \text{Gap}^{12m} \cdot \Delta r \tag{32}$$

where RSA^{12m} is the rate-sensitive asset balance repricing within 12 months, and RSL^{12m} is the corresponding liability balance.

The repricing gap captures only the *volume effect* of rate changes. Higher-order effects (yield curve non-parallelism, option costs, basis) require more sophisticated modelling.

5.2 EVE and Duration Gap

The Economic Value of Equity sensitivity:

$$\frac{\Delta\text{EVE}}{\text{EVE}} \approx -D_{\text{gap}} \cdot \frac{\Delta r}{1 + r} \tag{33}$$

where the duration gap is:

$$D_{\text{gap}} = D_A - \frac{L}{A} \cdot D_L \tag{34}$$

Positive D_{gap} means the bank is asset-sensitive in EVE terms (rising rates reduce EVE). Negative D_{gap} means liability-sensitive.

Proposition 1 (EVE-NII Trade-off)**.** *A bank cannot simultaneously minimize both NII sensitivity and EVE sensitivity to interest rate risk. Shortening asset duration increases NII in a rising rate environment but reduces EVE (less benefit from discounting liabilities at higher rates). This is the fundamental ALM trade-off.*

The optimal duration gap maximizes the utility function:

$$\max_{D_{\text{gap}}} \quad U(\Delta\text{NII}, \Delta\text{EVE}) = w_1 \cdot \mathbb{E}[\Delta\text{NII}] - w_2 \cdot \text{Var}(\Delta\text{NII}) - w_3 \cdot |\Delta\text{EVE}|^2 \tag{35}$$

subject to duration gap within the risk appetite band $[D^-, D^+]$.

5.3 Hedging the Duration Gap

If the current duration gap exceeds the target, the ALM desk uses interest rate swaps to close the gap:

- **Pay-fixed receiver IRS:** Converts a floating-rate liability into fixed-rate, increasing effective liability duration.
- **Receive-fixed payer IRS:** Converts a fixed-rate asset into floating, reducing effective asset duration.

The notional of swap required to close a duration gap of ΔD over a target balance sheet of A:

$$N_{\text{swap}} = \frac{\Delta D \cdot A}{D_{\text{swap}}} \tag{36}$$

where $D_{\text{swap}} = \text{DV01}_{\text{swap}}/\text{Price}_{\text{swap}}$.

5.4 Convexity and Prepayment Risk

Fixed-rate mortgages and callable bonds exhibit *negative convexity*: when rates fall, borrowers prepay (shortening asset duration when the bank least wants it) and when rates rise, borrowers extend (lengthening duration when the bank least wants that).

The effective duration and convexity:

$$P(r) \approx P(r_0) \left[1 - D_{\text{eff}} \cdot \Delta r + \frac{1}{2} C_{\text{eff}} \cdot (\Delta r)^2 \right] \tag{37}$$

For a pass-through MBS, $C_{\text{eff}} < 0$, creating a P&L concavity that increases with the size of the rate move.

A bank with a large MBS portfolio must hedge convexity separately using:

- Receiver swaptions (provide positive convexity when rates fall)
- CMS caps/floors
- Callable bond portfolios to offset MBS negative convexity

6 Fund Transfer Pricing (FTP)

6.1 Matched-Maturity FTP Architecture

The FTP framework transfers the cost of funding and the benefit of funding between the business lines and a central Treasury pool. Every asset originates a charge; every liability originates a credit:

$$\text{FTP_Charge}(k) = \underbrace{r^{\text{OIS}}(\tau_k)}_{\text{Funding cost}} + \underbrace{\lambda_k}_{\text{Liquidity premium}} + \underbrace{\psi_k}_{\text{Optionality premium}} \tag{38}$$

where:

- $r^{\text{OIS}}(\tau_k)$: OIS swap rate at the contractual maturity τ_k
- λ_k: liquidity premium representing the cost of term funding above OIS
- ψ_k: optionality premium for embedded options (prepayment, early withdrawal, commitment fee)

6.2 FTP and the Elimination of Cross-Subsidy

Without FTP, a bank using short-term cheap deposits to fund long-term loans appears to generate a large NII spread. In reality, it is accumulating unhedged rollover risk. FTP corrects this:

$$\text{NIM}_{\text{adjusted}}(k) = r_k^A - \text{FTP}(k) - \text{EL}_k \tag{39}$$

A loan desk with high gross NIM but high FTP charge (long-dated fixed-rate) may be less attractive than a shorter-duration product with lower gross NIM but a lower FTP charge.

6.3 Liquidity Premium Components

The liquidity premium λ contains:

$$\lambda = \lambda_{\text{term}} + \lambda_{\text{LCR}} + \lambda_{\text{NSFR}} + \lambda_{\text{CFP}} \tag{40}$$

- λ_{term}: term funding spread above OIS (market-observable from senior unsecured issuance)
- λ_{LCR}: cost of holding HQLA attributable to this liability class (high-runoff liabilities incur higher LCR charges)
- λ_{NSFR}: cost of providing stable funding to match RSF on the asset side
- λ_{CFP}: cost of maintaining Contingency Funding Plan reserves

6.4 Governance and Incentive Design

FTP design creates strong incentives. If FTP is set too low:

- Loan desks over-originate because the cost of funding is underpriced
- Balance sheet grows unsustainably

If FTP is set too high:

- Desks under-price deposits relative to true value
- Treasury retains excessive profit; business lines cross-subsidize Treasury

FTP rates must be reviewed at minimum quarterly by the ALM Committee, with Treasury as the internal market-maker.

7 RAROC and Risk-Adjusted Pricing

7.1 RAROC Framework

Risk-Adjusted Return on Capital (RAROC) is the primary metric for comparing across business lines with different risk profiles:

$$\mathrm{RAROC}(k) = \frac{(r_k^A - r_k^{\mathrm{FTP}}) \cdot A_k - \mathrm{EL}_k \cdot A_k - \mathrm{OpCost}_k}{\mathrm{EC}_k} \tag{41}$$

Definition 2 (RAROC Hurdle Rate)**.** *The hurdle rate h is typically the bank's cost of equity, estimated via CAPM:*

$$h = r_f + \beta \cdot (r_m - r_f) = 4.5\% + 1.2 \times 5.5\% = 11.1\% \tag{42}$$

for a money-center bank with $\beta \approx 1.2$. Any business line with RAROC $< h$ destroys shareholder value and should be restructured or exited.

7.2 Loan Pricing Using RAROC

The minimum loan rate r^* for a new loan that achieves the hurdle RAROC:

$$r^* = r^{\mathrm{FTP}}(\tau) + \lambda^{\mathrm{FTP}} + \mathrm{EL} + \frac{h \cdot \mathrm{EC}}{A} + \mathrm{OpCost} \tag{43}$$

Example: 5-year commercial loan

$$r^{\mathrm{OIS}}(5Y) = 4.40\% \tag{44}$$
$$\lambda^{\mathrm{FTP}}(5Y) = 0.45\% \tag{45}$$
$$\mathrm{EL} = \mathrm{PD} \cdot \mathrm{LGD} = 1.5\% \times 45\% = 0.675\% \tag{46}$$
$$\frac{h \cdot \mathrm{EC}}{A} = 11.1\% \times 8\% = 0.888\% \tag{47}$$
$$\mathrm{OpCost} = 0.20\% \tag{48}$$
$$r^* \approx 6.61\% \tag{49}$$

Any loan priced above 6.61% is value-creating. Below this rate, the bank is lending at a negative RAROC.

7.3 Capital Allocation and Performance Measurement

Economic capital is allocated to business lines proportional to their risk contribution. The allocation must satisfy:

$$\sum_k \mathrm{EC}_k = \mathrm{EC}_{\text{portfolio}} \tag{50}$$

Using Euler's theorem on homogeneous risk measures (VaR is 1-homogeneous):

$$\mathrm{EC}_k = w_k \cdot \frac{\partial \mathrm{VaR}_\alpha}{\partial w_k} \tag{51}$$

where w_k is the exposure weight of business line k. This ensures the diversification benefit is properly distributed.

8 Stress Testing Integration

8.1 From DFAST to Balance Sheet Strategy

DFAST stress testing is not merely a regulatory exercise — it is a strategic planning tool. The severely adverse CET1 trough determines:

- **Maximum loan growth capacity**: if the bank originates ΔA in new loans before DFAST, the stressed loss on those loans reduces the CET1 trough by approximately $\mathrm{EL}_{\text{stress}} \cdot \Delta A/\mathrm{RWA}$.
- **Maximum dividend capacity**: any dividend paid reduces starting CET1 and directly lowers the trough.
- **Buyback capacity**: capital returned via buybacks above the buffer could cause the trough to breach the internal floor.

The balance sheet optimization explicitly incorporates the DFAST constraint (equation (16)) as a planning constraint.

8.2 Reverse Stress Testing

Reverse stress testing identifies the balance sheet composition that would cause the bank to breach regulatory minima. The reverse stress problem:

$$\min_{\text{scenario}} \text{CET1_trough(scenario)} \quad \text{s.t.} \quad \text{Scenario plausible under risk model} \tag{52}$$

Key reverse stress scenarios:

1. **Credit concentration**: a sector-concentrated credit shock (e.g., CRE collapse of -55%) combined with deposit run causes simultaneous asset loss and liability instability.
2. **Duration trap**: a sharp rate rise (+300bps rapidly) reduces EVE to near-zero while simultaneously increasing funding costs on short-dated liabilities (SVB-type scenario).

3. **Liquidity-solvency spiral**: an LCR breach forces fire-sale of HQLA, which depresses prices of the HQLA assets themselves (pro-cyclical).

Each reverse stress scenario generates a specific "distance to failure" metric:

$$D_{\text{failure}} = \frac{\text{CET1_current} - \text{CET1_at_failure}}{\text{CET1_1-year-vol}} \tag{53}$$

9 Technology, Calculators, and Infrastructure

9.1 Required Calculator Stack

Balance sheet optimization requires an integrated stack of quantitative engines. The minimum production set:

Table 2: Core Calculator Stack for Balance Sheet Optimization

Calculator	Purpose	Key Output
RWA / Capital Engine	SA and IMA RWA; CET1/Tier1/Total ratios	Capital ratios
VaR / SVaR Engine	IMA market risk capital	10-day 99% VaR, SVaR
FRTB-SA Engine	SA sensitivities; curvature; DRC	FRTB-SA capital
IFRS 9 ECL Engine	Loan loss provision; staging	ECL coverage ratio
DFAST Engine	9-quarter CET1 projection	CET1 trough
ALM Engine	NII/EVE sensitivity; repricing gap	Duration gap; NIM
FTP Engine	Matched-maturity funding charges	Desk P&L (FTP-adjusted)
Collateral / SIMM	VM/IM; liquidity stress	IM requirement; stress call
XVA Suite	CVA/DVA/FVA/MVA/ColVA/KVA	XVA charge per trade
RAROC Calculator	Risk-adjusted pricing; hurdle test	Minimum loan rate
LCR / NSFR Calculator	Liquidity ratio compliance	LCR, NSFR ratios
Concentration Monitor	Single-name/sector/geo limits	Breach flags, HHI

9.2 Data Architecture

The calculators require a consistent, real-time data layer:

- **Positions layer**: real-time feed from OMS and loan systems; all instruments mapped via instrument master (ISIN/CUSIP/product type)
- **Market data layer**: live prices, rates, vol surfaces, credit spreads; GBM simulation for real-time VaR (Apex uses 500ms tick interval)
- **Reference data**: CSA terms, counterparty credit ratings, FATF grey-list, PEP database, ISDA SIMM risk weights
- **Event log**: append-only audit trail of all balance sheet changes, capital actions, and model outputs (SQLite in Apex implementation)
- **Regulatory parameters**: Fed-published DFAST scenarios, Basel risk weights, LCR runoff rates (updated annually or upon regulatory change)

9.3 Computational Requirements

Table 3: Computational Requirements by Calculator

Calculator	Frequency	Latency	Compute	Memory
VaR MC (2,000 paths, 11 instruments)	Daily post-close	50ms	1 CPU	50MB
FRTB-SA (delta/vega/curvature)	Daily	200ms	4 CPU	200MB
DFAST (9Q, 3 scenarios)	Quarterly	5–30s	8 CPU	1GB
XVA (1,000 paths, 50 trades)	On trade	2–10s	16 CPU	4GB
SIMM (CRIF-based, all CSAs)	Daily	100ms	2 CPU	100MB
LCR / NSFR	Daily	500ms	2 CPU	200MB
ALM / NII sensitivity	Monthly	1s	2 CPU	100MB
IFRS 9 ECL (50 obligors)	Quarterly	5ms	1 CPU	20MB

For production-scale balance sheet optimization (850+ obligors, 10,000+ derivative trades, full FRTB-IMA), the compute requirements scale to:

- XVA: 64–128 GPU cores for real-time Monte Carlo (AAD sensitivities)
- FRTB IMA: historical simulation on 10-year daily data for 500+ risk factors
- DFAST: cluster parallelization across 3 scenarios $\times$ 9 quarters

9.4 Dashboards and Reporting

A best-practice balance sheet optimization dashboard suite consists of:

1. Capital Dashboard CET1/Tier1/Total/Leverage ratios vs. regulatory minimums and internal targets. RWA decomposition by risk type (credit, market, operational). Capital headroom to constraint breaches. Trajectory over prior 8 quarters.

2. Liquidity Dashboard LCR and NSFR ratios real-time (intraday updates for LCR). HQLA composition by Level (1/2A/2B). 30-day net cash outflow waterfall. Intraday liquidity utilization vs. credit line.

3. ALM / Rate Risk Dashboard NII sensitivity bar chart ($\pm$100bps, $\pm$200bps, instantaneous). EVE sensitivity. Repricing gap schedule (7 buckets, asset vs. liability bars). Duration gap trend vs. risk appetite band. Warning indicators (SVB-style flag if EVE sensitivity exceeds –15%).

4. DFAST / Stress Dashboard 9-quarter CET1 trajectory under all 3 scenarios (Plotly chart with min CET1 floor and internal risk appetite band). Loss attribution waterfall (credit, market, operational, PPNR). Distance to regulatory minimum.

5. FTP / RAROC Dashboard Desk-level FTP-adjusted P&L ranking. RAROC vs. hurdle rate by business line. New origination pricing heat-map. FTP curve visualization (OIS + liquidity premium by tenor).

6. XVA Dashboard CVA, DVA, FVA, MVA, ColVA, KVA by counterparty and trade. Live WebSocket updates on new trade submissions. XVA P&L explain (delta, vega, credit) vs. prior day.

7. Boardroom / Scenario Dashboard Real-time agent-driven scenario narrative. Scenario engine with live risk factor injection. Model registry and MDD governance status.

10 How Balance Sheet Optimization Generates Revenue

10.1 Revenue Sources by Lever

Balance sheet optimization generates value through several distinct mechanisms:

10.1.1 Capital Efficiency: Releasing Trapped Capital

When RWA can be reduced without sacrificing income, the freed capital can be redeployed into higher-RAROC activities or returned to shareholders.

Example – Loan portfolio restructuring: A bank holds \$10B of low-grade CRE loans (RW = 100%, generating 6% yield). These can be replaced with investment-grade corporate bonds (RW = 20%, yielding 5.5%):

$$\text{RWA reduction} = (100\% - 20\%) \times \$10B = \$8B \tag{54}$$

$$\text{CET1 freed} = 7\% \times \$8B = \$560M \tag{55}$$

$$\text{NII sacrifice} = (6\% - 5.5\%) \times \$10B = \$50M/\text{yr} \tag{56}$$

$$\text{Return on freed capital} = \$50M/\$560M = 8.9\% \text{ cost} \tag{57}$$

If the freed CET1 can be deployed into a business generating RAROC > 8.9%, the swap is value-creating.

10.1.2 NII Optimization: Positioning the Duration Gap

A bank that correctly positions its duration gap before a rate cycle captures NII uplift:

Example – Rate rise scenario: Starting NII \$40B/year, duration gap +0.41yr. Rate rises 200bps: ΔNII = +\$6.8B/yr (Apex ALM model). This is a structural P&L benefit worth capturing by avoiding over-hedging. A bank with zero duration gap would earn \$0 uplift; correctly calibrating the gap to +0.41yr (consistent with risk appetite) generates \$6.8B incremental NII under a +200bps scenario.

Conversely, if rates had fallen 200bps, the bank would have lost \$5.9B NII. The asymmetric positioning (positive gap in an asset-sensitive bank) reflects the rate environment view embedded in the ALM strategy.

10.1.3 Liquidity Premium Capture

Treasury can earn the term liquidity premium by issuing short-term paper and lending to the asset book at FTP rates that include the term premium. This "carry" between short funding and long assets is a persistent revenue source in a normally-sloped yield curve:

$$\text{Carry} = r^{\text{FTP}}(10Y) - r^{\text{FTP}}(3M) = 4.55\% - 5.15\% = -60\text{bps} \tag{58}$$

In the current inverted curve, the carry is negative, making it advantageous to issue long-term and hold short-dated assets — the opposite of the normal regime.

10.1.4 XVA Optimization: KVA and FVA Pricing

The bank can generate revenue by correctly pricing XVA charges into derivatives trades at inception. If the XVA charge is not priced, the bank delivers P&L upfront but accumulates a long-dated liability as the XVA is realized.

KVA (Capital Valuation Adjustment): The cost of holding regulatory capital against the trade over its lifetime:

$$\text{KVA} = h \cdot \int_0^T \mathbb{E}[\text{EC}(t)] \cdot \text{df}(t)\, dt \tag{59}$$

where h is the hurdle rate and $\text{EC}(t)$ is expected economic capital at time t. KVA is typically 5–15bps on notional for a 5-year IRS.

Pricing KVA into derivatives trades means the desk charges the client the full cost of the trade, including the capital cost — generating an immediate revenue recognition that compensates for the long-dated capital consumption.

10.1.5 Regulatory Arbitrage: Cleared vs. Bilateral

Clearing eligible derivatives via a CCP (LCH, CME) attracts lower SA-CCR and IM requirements:

$$\text{IM (bilateral SIMM)} \approx \$70M \tag{60}$$

$$\text{IM (LCH cleared)} \approx \$25M \tag{61}$$

$$\text{IM savings} = \$45M \tag{62}$$

At a funding cost of 5.25%, the IM saving generates an annual FVA/MVA saving of $\$45M \times 5.25\% = \$2.4M$/yr per portfolio of this size. For a large derivatives book (\$500B notional), clearing-driven IM reduction can generate \$50–150M/yr in FVA/MVA savings.

10.1.6 HQLA Yield Optimization

Within the LCR/NSFR constraints, optimizing the HQLA portfolio composition generates incremental yield:

$$\text{HQLA yield gain} = (r^{\text{optimal}} - r^{\text{cash}}) \times \text{HQLA balance} = (4.65\% - 0\%) \times \$45B = \$2.1B/\text{yr} \tag{63}$$

Replacing non-earning excess reserves with a duration-matched portfolio of Treasuries and Agency bonds earns 4.5–4.7% on \$45B = ∼\$2.1B/yr with no reduction in LCR compliance.

10.2 Summary: Revenue Impact of Balance Sheet Optimization

Table 4: Estimated Annual Revenue Impact (Apex Scale, \$285B Balance Sheet)

Optimization Lever	Annual P&L Impact	Key Model
HQLA yield optimization	+\$2.1B	ALM
NII duration positioning (\$+200bps scenario)	+\$6.8B	ALM
Clearing-driven IM reduction (MVA/FVA saving)	+\$50–150M	SIMM
KVA pricing into new derivatives	+\$100–250M	XV
CET1 release via lower-RWA assets	+\$200–500M (redeployed)	Capita
Loan pricing discipline (RAROC floor)	Prevents erosion of 50–100bps NIM	
FTP governance (correct term premium)	Eliminates \$0.5–1.5B cross-subsidy	FTP

The cumulative revenue uplift from a mature balance sheet optimization program relative to an unoptimized baseline is typically 15–40bps on total assets, or \$0.4–1.1B/yr at Apex scale. At JPMorgan scale (\$3.9T balance sheet), the program generates \$3–8B/yr in incremental returns.

11 Governance and Model Risk

11.1 SR 11-7 Governance for Optimization Models

Every quantitative model used in balance sheet optimization is subject to SR 11-7 model risk governance:

- **Model Development Document (MDD):** Each calculator (VaR, DFAST, ALM, FTP, SIMM, XVA) requires a complete MDD including theoretical basis, implementation, validation, limitations, use authorization, and open findings.
- **Independent validation:** All Tier 1 models must be validated by a Model Validation Officer independent of the model developer.
- **Ongoing monitoring:** Model KRIs (key risk indicators) tracked monthly: VaR backtesting exception rate, SIMM vs. actual IM reconciliation, DFAST vs. actual CET1 variance, NII model vs. actual NII error.
- **Use authorization:** Each model's Section 7 specifies authorized uses, prohibited uses, and the approval chain.

11.2 ALM Committee Governance

The ALM Committee is the governing body for balance sheet optimization decisions. Membership:

- CRO (Chair)
- CFO / Treasurer
- Head of Trading (business line representative)
- Head of Commercial Banking
- Head of Model Validation (advisory)
- Chief Risk Economist (macro forecasting)

The Committee meets monthly and reviews: NII sensitivity, EVE sensitivity, duration gap vs. target, LCR/NSFR ratios, FTP rates, and any proposed balance sheet transactions above materiality thresholds.

11.3 The Three Lines of Defence

1. **First line (Treasury / business lines):** Runs the ALM engine, FTP, and RAROC calculations. Originates hedging recommendations.
2. **Second line (Risk / Model Validation):** Independent review of ALM models and FTP rates. Validates DFAST projections. Monitors model KRIs.
3. **Third line (Internal Audit):** Annual audit of the balance sheet optimization program; testing of governance controls, data integrity, and regulatory compliance.

12 Conclusion

Balance sheet optimization is the highest-leverage quantitative discipline in banking. Its scope encompasses capital structure, liquidity management, interest rate risk, pricing discipline, and regulatory strategy. It is not a single model or a single optimization problem — it is an integrated system of models, calculators, governance processes, and strategic decisions that operates continuously over the life of the bank.

The key insight is that the regulatory constraints (capital ratios, LCR, NSFR, DFAST) are not merely compliance requirements — they are *priced inputs* to the optimization. A bank that treats them as binding constraints to be minimally satisfied leaves significant value on the table. A bank that treats them as the efficient frontier of the feasible set, and optimizes within it, can generate several hundred basis points of incremental RORWA relative to peers.

At Apex Global Bank, the infrastructure described in this document — the ALM engine, FTP engine, DFAST engine, XVA suite, SIMM collateral engine, and regulatory capital engine — provides the quantitative foundation for this optimization program. The SR 11-7 model governance framework ensures that the outputs of these models are reliable, independently validated, and used only within their authorized scope.

The next frontier is dynamic balance sheet optimization: replacing the current static quarterly cycle with a continuous, model-driven optimization that updates recommendations in real-time as market conditions change. This requires AAD (Adjoint Algorithmic Differentiation) sensitivities across all models, GPU-accelerated Monte Carlo, and reinforcement learning for the hedging decision layer — capabilities that are commercially available and being deployed at G-SIBs today.

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